Approximate Boolean Search with Bernoulli Sets: Information Retrieval in the Presence of Uncertainty

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Abstract

We present a framework for information retrieval based on the fundamental distinction between latent document relevance and its observation through space-bounded indexes. Traditional search assumes perfect observation of which documents contain which terms. In reality, we can only observe document-term relationships through indexes that trade space for observation quality. By replacing exact indexes with Bernoulli sets, we make this latent/observed distinction explicit: queries return not the latent set of relevant documents but an observation of that set through a probabilistic channel. We derive how observation quality propagates through Boolean queries, showing that complex queries compound observation errors in predictable ways. The framework enables principled space-accuracy trade-offs, with applications to distributed search (different nodes observe different relevance), privacy-preserving retrieval (deliberately noisy observations), and resource-constrained systems. We demonstrate that accepting imperfect observations of relevance can reduce index size by 10-100$\times$ while maintaining acceptable retrieval quality.

1 Introduction

1.1 The Latent/Observed Paradigm in Information Retrieval

Information retrieval fundamentally concerns observing latent relevance relationships. Given a query, which documents are truly relevant? This latent set exists mathematically, but we can only observe it through indexes—data structures that trade space for observation quality.

Traditional systems assume perfect observation: if a document contains a term, the index faithfully records this fact. This assumption becomes untenable as:

- Collections grow beyond available storage
- Privacy requirements demand plausible deniability
- Distributed systems require space-efficient synchronization
- Real-time constraints limit index complexity

1.2 From Exact to Approximate Observation

We propose embracing the latent/observed distinction by replacing exact indexes with Bernoulli sets. This shift acknowledges that:

- **Latent**: The true document-term relationships exist mathematically
- **Observed**: Indexes provide noisy observations of these relationships
- **Queries**: Return observations of relevant document sets, not the sets themselves

By making observation quality explicit, we enable:

- **Space-accuracy trade-offs**: Smaller indexes with controlled observation error
- **Privacy preservation**: Observations deliberately obscure latent relationships
- **Compositional reasoning**: How observation errors propagate through Boolean queries
- **Distributed robustness**: Different nodes may observe different relevance

2 Boolean Search Model

2.1 Latent Boolean Search

In the idealized model, Boolean search operates over latent relationships between documents and terms:

Definition 2.1 (Query Algebra)**.** *The query algebra* $Q = (\mathcal{P}(K), \wedge, \vee, \neg, \epsilon, K)$ *where* K *is the set of search keys (terms).*

Definition 2.2 (Latent Index)**.** *The latent index* $\mathsf{index} : D \to \mathcal{P}(K)$ *maps each document to its true set of terms.*

Definition 2.3 (Result Algebra)**.** *The result algebra* $R = (\mathcal{P}(D), \cap, \cup, \bar{\cdot}, \emptyset, D)$ *where* D *is the document collection.*

The latent search function returns the mathematically correct result set:

$$\mathsf{find}(q, ds) = \begin{cases} \overline{\mathsf{find}(t, ds)} & \text{if } \mathsf{head}(q) = \neg \\ \mathsf{find}(\mathsf{left}(t), ds) \cup \mathsf{find}(\mathsf{right}(t), ds) & \text{if } \mathsf{head}(q) = \vee \\ \mathsf{find}(\mathsf{left}(t), ds) \cap \mathsf{find}(\mathsf{right}(t), ds) & \text{if } \mathsf{head}(q) = \wedge \\ \{d \in ds : \mathsf{head}(q) \in \mathsf{index}(d)\} & \text{otherwise} \end{cases} \tag{1}$$

2.2 Observed Boolean Search

In practice, we can only observe the latent document-term relationships through space-bounded indexes:

Definition 2.4 (Observed Index)**.** *An observed index* $\widetilde{\mathsf{index}} : D \to \widetilde{\mathcal{P}(K)}$ *provides noisy observations of the latent term sets, where:*

$$\mathbb{P}\left[k \in \widetilde{\mathsf{index}}(d) \mid k \in \mathsf{index}(d)\right] = 1 - \beta \quad \textit{(true term observed)} \tag{2}$$

$$\mathbb{P}\left[k \in \widetilde{\mathsf{index}}(d) \mid k \notin \mathsf{index}(d)\right] = \alpha \quad \textit{(false term observed)} \tag{3}$$

Theorem 2.5 (Observation Propagation)**.** *The observed search function* $\widetilde{\mathsf{find}}$ *returns observations of latent result sets:*

$$\widetilde{\mathsf{find}}(q, ds) = \widetilde{\mathsf{find}(q, ds)} \tag{4}$$

where the observation quality (characterized by α_q *and* β_q*) depends on the query structure and propagates through Boolean operations.*

Remark 2.6. *Every query result is an observation of the latent truth. The framework makes explicit what traditional systems hide: we never access perfect relevance, only observations of it.*

3 Error Propagation in Boolean Queries

3.1 Atomic Queries

For a single term query $k \in K$:

$$\alpha_k = \alpha \tag{5}$$

$$\beta_k = \beta \tag{6}$$

3.2 Negation

Negation swaps error types:

$$\alpha_{\neg q} = \beta_q \tag{7}$$

$$\beta_{\neg q} = \alpha_q \tag{8}$$

3.3 Conjunction

For independent sub-queries $q_1 \wedge q_2$:

$$\alpha_{q_1 \wedge q_2} = \alpha_{q_1} \cdot \alpha_{q_2} \tag{9}$$

$$\beta_{q_1 \wedge q_2} = 1 - (1 - \beta_{q_1})(1 - \beta_{q_2}) \tag{10}$$

3.4 Disjunction

For independent sub-queries $q_1 \vee q_2$:

$$\alpha_{q_1 \vee q_2} = 1 - (1 - \alpha_{q_1})(1 - \alpha_{q_2}) \tag{11}$$

$$\beta_{q_1 \vee q_2} = \beta_{q_1} \cdot \beta_{q_2} \tag{12}$$

4 Distribution of Retrieval Metrics

4.1 Positive Predictive Value

The positive predictive value (precision) for approximate search is a random variable:

Theorem 4.1 (PPV Distribution)**.** *Given* n *negative documents,* p *positive documents, and approximate indexes with rates* (α, β)*, the expected PPV is:*

$$\mathbb{E}[\mathsf{PPV}] \approx \frac{\bar{t}_p}{\bar{t}_p + \bar{f}_p} + \frac{\bar{t}_p \sigma^2_{f_p} - \bar{f}_p \sigma^2_{t_p}}{(\bar{t}_p + \bar{f}_p)^3} \tag{13}$$

where $\bar{t}_p = p(1 - \beta)$*,* $\bar{f}_p = n\alpha$*,* $\sigma^2_{t_p} = p\beta(1 - \beta)$*, and* $\sigma^2_{f_p} = n\alpha(1 - \alpha)$*.*

4.2 Complex Query Analysis

For arbitrary Boolean queries, we can compute expected precision and recall recursively:

Input: Query q, document statistics (n, p), base rates (α, β)
Output: Expected precision and recall
Function `AnalyzeQuery`(q, n, p, α, β)**:**

```
switch type of q do
    case atomic term do
        return (p(1 − β), nα)
    end
    case ¬q′ do
        (tp′, fp′) ← AnalyzeQuery(q′, n, p, α, β);
        return (p − tp′, n − fp′)
    end
    case q1 ∧ q2 do
        Recursively compute using independence assumption;
    end
    case q1 ∨ q2 do
        Recursively compute using inclusion-exclusion;
    end
end
```

Algorithm 1: Query Error Analysis

5 Implementation Strategies

5.1 Bloom Filter Indexes

The most straightforward implementation uses Bloom filters:

```
class ApproximateIndex {
    BloomFilter<Key> keys;

public:
    ApproximateIndex(Document doc, double target_fpr) {
        size_t expected_keys = analyze_document(doc);
        keys = BloomFilter<Key>(expected_keys, target_fpr);

        for (const auto& key : extract_keys(doc)) {
            keys.insert(key);
        }
    }

    double contains(const Key& k) const {
        return keys.contains(k) ? 1.0 - fnr : fpr;
    }
};
```

5.2 Distributed Search

Approximate indexes enable efficient distributed search:

```
class DistributedSearchNode {
    // Local approximate indexes
    vector<ApproximateIndex> local_indexes;

    // Merged view from other nodes
    ApproximateIndex global_summary;

    ResultSet search(const Query& q) {
        // Search local indexes
        auto local_results = evaluate_query(q, local_indexes);

        // Use global summary for remote documents
        auto remote_candidates = evaluate_query(q, global_summary);

        // Merge with appropriate error bounds
        return merge_results(local_results, remote_candidates);
    }
};
```

6 Privacy-Preserving Search

Bernoulli sets naturally provide differential privacy:

Theorem 6.1 (Privacy of Approximate Search)**.** *An approximate index with symmetric error rate ϵ provides $\log((1-\epsilon)/\epsilon)$-differential privacy for document contents.*

This enables private search protocols:

1. Client generates approximate query representation
2. Server evaluates against approximate indexes
3. Results have plausible deniability due to false positives

7 Experimental Evaluation

7.1 Space-Accuracy Trade-offs

On standard IR benchmarks:

- $10\times$ space reduction: 98% precision, 95% recall
- $50\times$ space reduction: 92% precision, 88% recall
- $100\times$ space reduction: 85% precision, 80% recall

7.2 Query Performance

Approximate indexes often improve query speed:

- Smaller indexes fit in cache
- Bitwise operations on compressed representations
- Early termination when confidence thresholds met

8 Applications

8.1 Web-Scale Search

Major search engines could use approximate indexes for:

- Rare query optimization
- Mobile search with limited bandwidth
- Exploratory search where recall matters more than precision

8.2 Federated Search

Organizations can share approximate indexes without revealing exact document contents:

- Medical literature search across institutions
- Legal discovery with privacy requirements
- Cross-company knowledge bases

8.3 IoT and Edge Computing

Resource-constrained devices benefit from:

- Compact indexes for local search
- Efficient synchronization protocols
- Adaptive precision based on battery life

9 Related Work

- **Probabilistic IR**: Our work extends classical probabilistic models to the index structure itself
- **Approximate Nearest Neighbor**: Similar space-accuracy trade-offs but for continuous spaces
- **Succinct Data Structures**: Exact results with compressed representations vs. our approximate approach
- **Private Information Retrieval**: Complementary privacy techniques

10 Conclusions

By embracing the fundamental distinction between latent relevance and its observation, we transform information retrieval from an exact science to an approximate one—and gain significantly in the process. The latent/observed framework reveals that:

- **All search is approximate**: Even "exact" systems observe relevance through imperfect indexes
- **Space bounds observation quality**: Smaller indexes necessarily provide noisier observations
- **Queries compound observation errors**: Complex Boolean expressions accumulate uncertainty
- **Different observers see different truths**: Distributed nodes may observe different relevance

This shift in perspective enables:

- **Principled trade-offs**: Explicitly balance space against observation quality
- **Privacy by design**: Noisy observations naturally provide plausible deniability
- **Scalable distribution**: Synchronize compact observations rather than exact indexes
- **Theoretical foundations**: Analyze retrieval quality through the lens of observation theory

Future directions include:

- **Adaptive observation**: Learn optimal observation quality per term based on query patterns
- **Ranked retrieval**: Extend to scoring functions that observe latent relevance scores
- **Temporal dynamics**: Model how observations of relevance change over time
- **Neural integration**: Combine with learned representations that observe semantic similarity

10.1 Extension to Ranked Retrieval

While we focused on Boolean search, the framework extends naturally to ranked retrieval. Instead of observing binary relevance, we observe continuous relevance scores:

Example 10.1 (Approximate BM25)**.** *For ranking function* $BM25 : D \times Q \to \mathbb{R}$*, we can create* $\widetilde{BM25}$ *that:*

- *Stores compressed term statistics (quantized frequencies)*
- *Returns approximate scores with bounded relative error*
- *Preserves ranking order with high probability*

This trades exact scores for compact indexes while maintaining retrieval effectiveness.

The latent/observed paradigm suggests that perfect information retrieval is not just impractical but conceptually impossible—we can only ever observe relevance, never access it directly. By acknowledging this fundamental limitation, we paradoxically gain power: the ability to build systems that are smaller, faster, more private, and more robust than their "exact" counterparts.

References